

COST ANALYSIS & PRODUCTION ANALYSIS

Concepts of Production, production function with one variable input - Law of Variable Proportion, Laws of returns to scale, Indifference Curves, ISO-Quants & ISO-Cost line, Economies of scale, Diseconomies of scale. Types of cost, Cost curves, Cost – Output Relationship in the short run and in the long run, Long- Run Average Cost (LAC) curve

Break Even Analysis—Meaning, Assumptions, Determination of BEA, Limitations, Margin of safety, Uses of BEA In Managerial decisions (Theory and simple Problems).

INTRODUCTION OR PRODUCTION CONCEPTS:

Production can be defined as an organized activity of transforming physical inputs into output which will satisfy the products needs of the society.

or

Production is an act of creating value that satisfies the wants of the individuals.

Production refers to the transformation of inputs or resources into outputs or goods and services.

Production is a process in which economic resources or inputs (composed of natural resources like labour, land and capital equipment) are combined by entrepreneurs to create economic goods and services (outputs or products).

PRODUCTION ANALYSIS:

Production Analysis refers to analyzing the inputs or resources that are used to produce a firm's final product. It defines the relationships between the prices of the commodities and productive factors on one hand and the quantities of these commodities and productive factors that are produced on the other hand.

Production analysis basically is concerned with the analysis in which the resources such as land, labor, and capital are employed to produce a firm's final product. To produce these goods the basic inputs are classified into two divisions

Inputs : Fixed inputs and Variable inputs

The factors of production that carry out the production is called inputs.

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Fixed inputs	Variable inputs
☐ Remain the same in the short period . remain the same. Cost ☐ Examples:- Building, Land etc ☐ (In the long run fixed inputs are become varies)	☐ In the long run all factors of production are varies according to the volume of outputs. ☐ The cost of variable inputs is called Variable Cost ☐ Example:- Raw materials, labour, etc

What is Production Function?

Production Function signifies the technical relationship between the physical inputs and physical outputs of the firm for given production technology.

The basic relationship between the factors of production and the output is referred to as a Production Function.

The firm's production function for a particular good (q) shows the maximum amount of the good that can be produced using alternative combinations of capital (K) and labor (L)

The production function expresses a functional relationship between physical inputs and physical outputs of a firm at any particular time period. The output is thus a function of inputs. Mathematically production function can be written as

$$Q = f(L_1, L_2, C, O, T)$$

Importance:

1. When inputs are specified in physical units, production function helps to estimate the level of production.
2. It becomes is equates when different combinations of inputs yield the same level of output.
3. It indicates the manner in which the firm can substitute on input for another without altering the total output.
4. When price is taken into consideration, the production function helps to select the least combination of inputs for the desired output.
5. It considers two types" input-output relationships namely „law of variable proportions" and „law of returns to scale". Law of variable propositions explains the pattern of output in the short-run as the units of variable inputs are increased to increase the output. On the other hand law of returns to scale explains the pattern of output in the long run as all the units of inputs are increased.

6. The production function explains the maximum quantity of output, which can be produced, from any chosen quantities of various inputs or the minimum quantities of various inputs that are required to produce a given quantity of output.

Assumptions:

Production function has the following assumptions.

1. The production function is related to a particular period of time.
2. There is no change in technology.
3. The producer is using the best techniques available.
4. The factors of production are divisible.
5. Production function can be fitted to a short run or to long run.

Types of production function:-

These two types of relationships have been explained in the form of laws.

- i) Law of variable proportions (short run production function)
- ii) Law of returns to scale (long run production function)

I. Law of variable proportions:

It is also called law of diminishing returns or law of returns to variable factor.

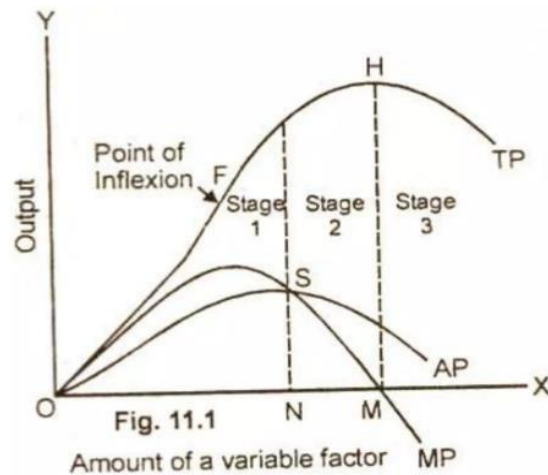
The law of variable proportions which is a new name given to old classical concept of “Law of diminishing returns has played a vital role in the modern economics theory. Assume that a firms production function consists of fixed quantities of all inputs (land, equipment, etc.) except labour which is a variable input when the firm expands output by employing more and more labour it alters the proportion between fixed and the variable inputs. The law can be stated as follows:

“When total output or production of a commodity is increased by adding units of a variable input while the quantities of other inputs are held constant, the increase in total production becomes after some point, smaller and smaller”

“If equal increments of one input are added, the inputs of other production services being held constant, beyond a certain point the resulting increments of product will decrease i.e. the marginal product will diminish”. (G. Stigler)

“As the proportion of one factor in a combination of factors is increased, after a point, first the marginal and then the average product of that factor will diminish”. (F. Benham)

The law of variable proportions refers to the behaviour of output as the quantity of one Factor is increased Keeping the quantity of other factors fixed and further it states that the marginal product and average product will eventually do cline. This law states three types of productivity an input factor – Total, average and marginal physical productivity.



Assumptions of the Law: The law is based upon the following assumptions:

- i) The state of technology remains constant. If there is any improvement in technology, the average and marginal output will not decrease but increase.
- ii) Only one factor of input is made variable and other factors are kept constant. This law does not apply to those cases where the factors must be used in rigidly fixed proportions.
- iii) All units of the variable factors are homogenous.

Three stages of law:

The behaviors of the Output when the varying quantity of one factor is combines with a fixed quantity of the other can be divided in to three district stages. The three stages can be better understood by following the table.

Table 11.1: Production Schedule

Units of Variable Input (Labour) (n)	Total Product (TP)	Average Product (AP) (TP_n/n)	Marginal Product (MP) ($TP_n - TP_{n-1}$)	
1	20	20	20	Stage I
2	50	25	30	
3	90	30	40	
4	120	30	30	
5	135	27	15	Stage II
6	144	24	9	
7	147	21	3	
8	148	18.5	1	
9	148	16.4	0	Stage III
10	145	14.5	- 3	

In the first stage, total product increases at an increasing rate. The marginal product in this stage increases at an increasing rate resulting in a greater increase in total product. The average product also increases. This stage continues up to the point where average product is

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equal to marginal product. The law of increasing returns is in operation at this stage. The law of diminishing returns starts operating from the second stage onwards. At the second stage total product increases only at a diminishing rate. The average product also declines.

The second stage comes to an end where total product becomes maximum and marginal product becomes zero. The marginal product becomes negative

In the third stage. So the total product also declines. The average product continues to decline.

We can sum up the above relationship thus when „A.P.“ is rising, “M. P.” rises more than “A. P.”; When „A. P.” is maximum and constant, „M. P.” becomes equal to „A. P.” when „A. P.” starts falling, „M. P.” falls faster than „A. P.” Thus, the total product, marginal product and average product pass through three phases, viz., increasing diminishing and negative returns stage. The law of variable proportion is nothing but the combination of the law of increasing and diminishing returns.

Total Product	Marginal Product	Average Product
Stage I First increases at increasing rate then at diminishing rate.	Increases in the beginning then reaches a maximum and begins to decrease.	First increases, continues to increase and becomes maximum.
Stage II Continues to increase at diminishing rate and becomes maximum.	Continues to diminish and becomes equal to zero.	Becomes equal to MP and then begins to diminish.
Stage III Diminishes	Becomes negative.	Continues to diminish but will always be greater than zero.

PRODUCTION FUNCTION WITH TWO VARIABLE INPUTS:

Refers to the relationship between the output resulting in production by varying the two or more inputs. There may be various technical possibilities of producing a given output by using different factor combinations. Which particular factor combination will be actually selected by the firm depends both on the technical possibilities of factor substitution as well as on the prices of the factors of production.

Production Function with two Variable Inputs explains the production behavior of the firm with all variable factors. We are restricting our analysis to two variable inputs because it simply allows us the scope for graphical analysis.\

II. Law of Returns of Scale:

The law of returns to scale explains the behavior of the total output in response to change in the scale of the firm, i.e., in response to a simultaneous to changes in the scale of the firm, i.e., in response to a simultaneous and proportional increase in all the inputs. More precisely, the Law of returns to scale explains how a simultaneous and proportionate increase in all the inputs affects the total output at its various levels.

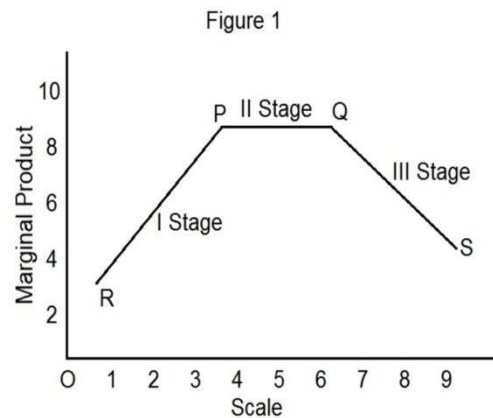
When a firm expands, its scale increases all its inputs proportionally, then technically there are three possibilities. (i) The total output may increase proportionately (ii) The total output may increase more than proportionately and (iii) The total output may increase less than proportionately.

Assumptions of Law of returns to scale:

- All inputs or factors of production are variable
- Production technology remains constant
- Product produce is measure in physical units

Unit	Scale of Production	Total Returns	Marginal Returns
1	1 labor + 2 Acres of Land	4	4 (Stage I - Increasing Returns)
2	2 Labor + 4 Acres of Land	10	6
3	3 Labor + 6 Acres of Land	18	8
4	4 Labor + 8 Acres of Land	28	10 (Stage II - Constant Returns)
5	5 Labor + 10 Acres of Land	38	10
6	6 Labor + 12 Acres of Land	48	10
7	7 Labor + 14 Acres of Land	56	8 (Stage III - Decreasing Returns)
8	8 Labor + 16 Acres of Land	62	6

Stage 1: Increasing Returns to Scale In figure, stage I represents increasing returns to scale. During this stage, the firm enjoys various internal and external economies such as technical economies, managerial economies and marketing economies. Economies simply mean advantages for the firm. Due to these economies, the firm realizes increasing returns to scale. This stage is also explained saying increasing returns in terms of



“increased efficiency” of labor and capital in the improved organization with the expanding scale of output and employment factor unit. It is referred to as the economy of organization in the earlier stages of production.

Increasing Return to Scale: If increase in the output is greater than the proportional increase in the inputs, it means increasing return to scale

Stage 2: Constant Returns to Scale

In figure, the stage II represents constant returns to scale. During this stage, the economies accrued during the first stage start vanishing and diseconomies arise. Diseconomies refers to the limiting factors for the firm’s expansion. Emergence of diseconomies is a natural process when a firm expands beyond certain stage. In the stage II, the economies and diseconomies of scale are exactly in balance over a particular range of output. When a firm is at constant returns to scale, an increase in all inputs leads to a proportionate increase in output but to an extent.

Constant returns to scale: If increase in the total output is proportional to the increase in input, it means constant returns to scale.

Stage 3: Diminishing Returns to Scale

In figure, the stage III represents diminishing returns or decreasing returns. This situation arises when a firm expands its operation even after the point of constant returns. Decreasing returns mean that increase in the total output is not proportionate according to the increase in the input. Because of this, the marginal output starts decreasing. Important factors that determine diminishing returns are managerial inefficiency and technical constraints.

Diminishing Returns to Scale: If increase in the output is less than proportional increase in the inputs, it means diminishing returns to scale.

ISOQUANTS:

The term Isoquants is derived from the words „iso“ and „quant“ – „Iso“ means equal and „quant“ implies quantity. Isoquant therefore, means equal quantity. A family of iso-product curves or isoquants or production difference curves can represent a production function with two variable inputs, which are substitutable for one another within limits.

Isoquants are the curves, which represent the different combinations of inputs producing a particular quantity of output. Any combination on the Isoquant represents the same level of output.

For a given output level firm's production becomes,

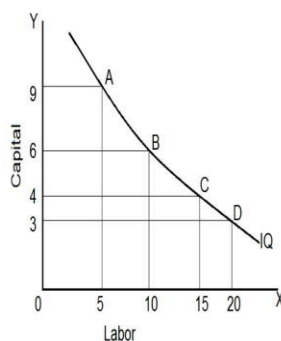
$$Q = f(L, K)$$

Where „Q“, is the units of output is a function of the quantity of two inputs „L“ and „K“.

Thus an Isoquant shows all possible combinations of two inputs, which are capable of producing equal or a given level of output. Since each combination yields same output, the producer becomes indifferent towards these combinations.

Assumptions:

1. There are only two factors of production, viz. labour and capital.
2. The two factors can substitute each other up to a certain limit
3. The shape of the Isoquant depends upon the extent of substitutability of the two inputs.
4. The technology is given over a period.



The above table is based on the assumption that only two factors of production, namely, Labor and Capital are used for producing 100 meters of cloth.

Combination A = $5L + 9K = 100$ meters of cloth

Combination B = $10L + 6K = 100$ meters of cloth

Combination C = $15L + 4K = 100$ meters of cloth

Combination D = $20L + 3K = 100$ meters of cloth

Marginal Rate of Technical Substitution:

Marginal rate of technical substitution in the theory of production is similar to the concept of marginal rate of substitution in the indifference curve analysis of consumer's demand. Marginal rate of technical substitution indicates the rate at which factors can be substituted at the margin without altering the level of output.

PROPERTIES OF ISOQUANTS

Property 1: An isoquant curve slopes downward, or is negatively sloped. This means that the same level of production only occurs when increasing units of input are offset with lesser

units of another input factor. This property falls in line with the principle of the Marginal Rate of Technical Substitution (MRTS). As an example, the same level of output could be achieved by a company when capital inputs increase, but labor inputs decrease.

Property 2: An isoquant curve, because of the MRTS effect, **is convex to its origin**. This indicates that factors of production may be substituted with one another. The increase in one factor, however, must still be used in conjunction with the decrease of another input factor.

Property 3: Isoquant curves cannot be tangent or intersect one another. Curves that intersect are incorrect and produce results that are invalid, as a common factor combination on each of the curves will reveal the same level of output, which is not possible.

Property 4: Isoquant curves in the **upper portions of the chart yield higher outputs**. This is because, at a higher curve, factors of production are more heavily employed. Either more capital or more labor input factors result in a greater level of production.

Property 5: An isoquant curve **should not touch the X or Y axis on the graph**. If it does, the rate of technical substitution is void, as it will indicate that one factor is responsible for producing the given level of output without the involvement of any other input factors.

Property 6: Isoquant curves **do not have to be parallel to one another**; the rate of technical substitution between factors may have variations.

Property 7: Isoquant curves are oval-shaped, allowing firms to determine the most efficient factors of production.

TYPES OF ISOQUANT:

Isoquants are differentiated on the basis of substitutability of the factors of production and they are as below

1. Linear Isoquant:

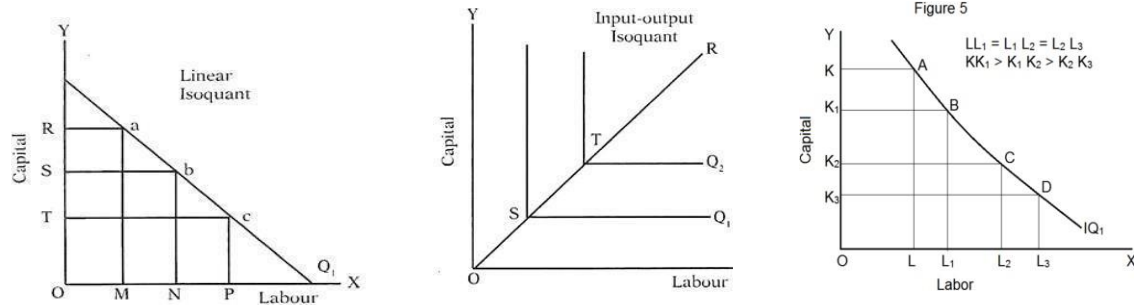
This type of isoquant are depicted by a straight line sloping downward from left to right, as shown in Figure-(a). It indicated a perfect and unlimited substitutability between two factors implying that the product may be produced even by using only capital or labour or by infinite combinations of the two factors.

2. Right angled Isoquant:

are L-shaped curve Figure-(b) and also known as Leontief isoquants. They assume a perfect complementary nature between factors implying zero substitutability. Factors are jointly used in a fixed proportion. It means that there is only one method of production to produce a commodity. Hence, to increase output, both factors are to be increased holding the proportion constant.

3. Convex Isoquant:

In this the inputs can be substituted but not perfectly. The curve of this type is convex to origin as we always consider for the isoquant curve.



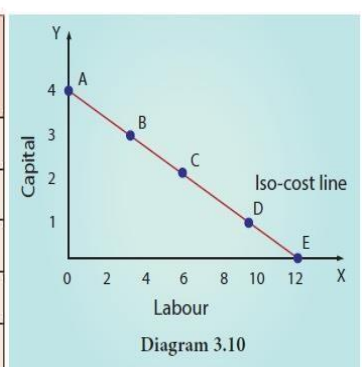
ISO-COST CURVE:

Isocost line shows all combinations of inputs which cost the same total amount. The use of the Isocost line pertains to cost-minimization in production, as opposed to utility-maximization. An Isocost line shows the maximum amount which a firm is willing to expend on production.

Slope of iso cost line: With the change in the factor prices the slope of iso cost line will change. If the price of labour falls the firm could buy more of labour and the line will shift away from the origin. The slope depends on the prices of factors of production and the amount of money which the firm spends on the factors. When the amount of money spent by the firm changes, the isocost line may shift but its slope remains the same. A change in factor price makes changes in the slope of isocost lines as shown in the figure.

Table 3.3 The Iso-cost

Combinations	Units of Capital Price = ₹30	Units of Labour Price = ₹10	Total Expenditure (in Rupees)
A	4	0	120
B	3	3	120
C	2	6	120
D	1	9	120
E	0	12	120

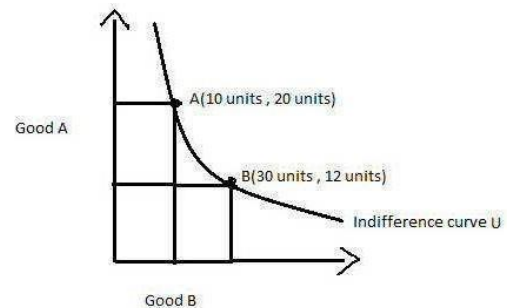


INDIFFERENCE CURVE: An indifference curve connects points on a graph representing different quantities of two goods, points between which a consumer is indifferent. An indifference curve is a graph showing combination of two goods that give the consumer equal

satisfaction and utility. Each point on an indifference curve indicates that a consumer is indifferent between the two and all points give him the same utility.

Graphically, the indifference curve is drawn as a downward sloping convex to the origin. The graph shows a combination of two goods that the consumer consumes. The above diagram shows the U indifference curve showing

bundles of goods A and B. To the consumer, bundle A and B are the same as both of them give him the equal satisfaction. In other words, point A gives as much utility as point B to the individual. The consumer will be satisfied at any point along the curve assuming that other things are constant.



LEAST COST COMBINATION FACTOR:

Producer's equilibrium or optimization occurs when he earns maximum profit with optimal combination of factors. A profit maximization firm faces two choices of optimal combination of factors

1. To Maximize its output for a given cost

2. To minimize its cost for given output

Thus the least cost combination of factors refers to a firm producing the largest volume of output from a given cost and producing a given level of output with the minimum cost when the factors are combined in an optimum manner.

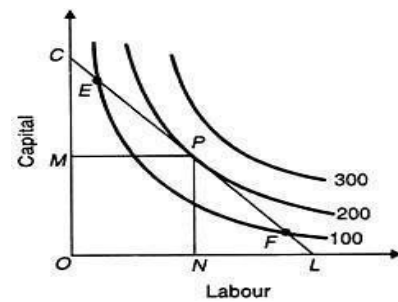
Assumptions made for to understand this concept are:

1. There are two factors, labour and capital.
2. All units of labour and capital are homogeneous.
3. The prices of units of labour (w) and that of capital (r) are given and constant.
4. The cost outlay is given.
5. The firm produces a single product.
6. The price of the product is given and constant.
7. The firm aims at profit maximisation.
8. There is perfect competition in the factor market.

1. To maximize output for given cost:

The firm maximizes its profits by maximizing its output, given its cost outlays, and the prices of the two factors. This analysis is based on the same assumptions, as given above From the

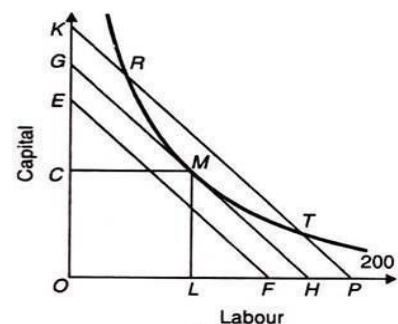
fig, the firm can reach the optimal factor combination level of maximum output by moving along the iso-cost line CL from either point E or F to point P. This movement involves no extra cost because the firm remains on the same iso-cost line the firm is maximising its output level of 200 units by employing the optimal combination of OM of capital and ON of labour, given its cost outlay CL.



But it cannot be at points E or F on the iso-cost line CL, since both points give a smaller quantity of output, being on the isoquant 100, than on the isoquant 200. The firm cannot attain a higher level of output such as isoquant 300 because of the cost constraint. The firm is in equilibrium at point P where the isoquant curve 200 is tangent to the iso-cost line CL. Thus the equilibrium point has to be P with optimal factor combination OM + ON.

2. To minimize cost for given output:

Given these assumptions, the point of least-cost combination of factors for a given level of output is where the isoquant curve is tangent to an iso-cost line. The iso-cost line GH is tangent to the isoquant 200 at point M. The firm employs the combination of OC of capital and OL of labour to produce 200 units of output at point M with the given cost-outlay GH.



At this point, the firm is minimising its cost for producing 200 units.

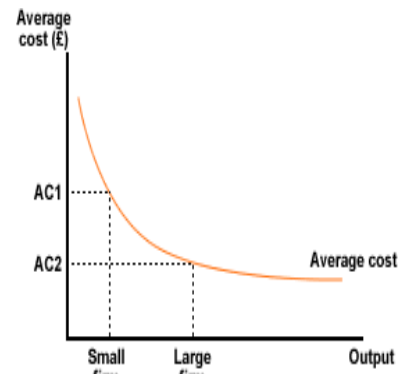
Any other combination on the isoquant 200, such as R or T, is on the higher iso-cost line KP which shows higher cost of production. The iso-cost line EF shows lower cost but output 200 cannot be attained with it. Therefore, the firm will choose the minimum cost point M which is the least-cost factor combination for producing 200 units of output.

M is thus the optimal combination for the firm. The point of tangency between the iso-cost line and the isoquant is an important first order condition but not a necessary condition for the producer's equilibrium.

ECONOMIES & DISECONOMIES OF SCALE: Economies of scale are defined as the cost advantages that an organization can achieve by expanding its production in the long run. In other words, these are the advantages of large scale production of the organization. The cost advantages are achieved in the form of lower average costs per unit.

It is a long term concept. Economies of scale are achieved when there is an increase in the sales of an organization. Economies of scale are the cost advantages that enterprises obtain due to their scale of operation, with cost per unit of output decreasing which causes scale increasing.

Figure illustrates that average cost falls as output increases, with the result that large firms may enjoy lower costs than smaller competitors. This competitive cost advantage allows large firms to have larger profit margins and have more options in pricing policy.



Reason for Economies of Scale:

1. Managerial - managers are on a fixed salary
2. Marketing - advertising, endorsements promotional events do not directly depend on quantity produced
3. Technical - machinery, buildings etc are paid for as a fixed amount
4. Bulk buying - remember it is the cost per unit of buying in bulk not the total cost (Great example is supermarkets and local shop)
5. Financial - similar in principle to buying in bulk but this time interest rates a more favorable.

The economies of scale are divided in to internal economies and external economies:

INTERNAL ECONOMIES:

As a firm increases its scale of production, the firm enjoys several economies named as internal economies. Basically, internal economies are those which are special to each firm. For example, one firm will enjoy the advantage of good management; the other may have the advantage of specialisation in the techniques of production and so on.

“Internal economies are those which are open to a single factory, or a single firm independently of the action of other firms. These result from an increase in the scale of output of a firm and cannot be achieved unless output increases.” Cairncross

Refer to real economies which arise from the expansion of the plant size of the organization. These economies arise from the growth of the organization itself.

A. Technical Economies of Scale: Technical economies have their influence on the size of the firm. Generally, these economies accrue to large firms which enjoy higher efficiency from capital goods or machinery. Bigger firms having more resources at their disposal are

able to install the most suitable machinery. Therefore, a firm producing on large scale can enjoy economies by the use of superior techniques.

Technical economies are of three kinds:

(i) Economies of Dimension:

A firm by increasing the scale of production can enjoy the technical economies. When a firm increases its scale of production, average cost of production falls but its average return will be more.

(ii) Economies of Linked Process:

A big firm can also enjoy the economies of linked process. A big firm carries all productive activities. These activities get economies. These linked activities save time and transport costs to the firm.

(iii) Economies of the Use of By-Products:

All the large sized firms are in a position to use its by-products and waste-material to produce another material and thus, supplement to their income. For instance, sugar industries make power, alcohol out of the molasses.

B. Marketing Economies of Scale:

When the scale of production of a firm is increased, it enjoys numerous selling or marketing economies. In the marketing economies, we include advertisement economies, opening up of show rooms, appointment of sole distributors etc. Moreover, a large firm can conduct its own research to effect improvement in the quality of the product and to reduce the cost of production. The other economies of scale are advertising economies, economies from special arrangements with exclusive dealers. In this way, all these acts lead to economies of large scale production.

C. Labor Economies of Scale:

As the scale of production is expanded they accrue many labour economies, like new inventions, specialization, time saving production etc. A large firm employs large number of workers. Each worker is given the kind of job he is fit for. The personnel officer evaluates the working efficiency of the labour if possible. Workers are skilled in their operations which save production, time and simultaneously encourage new ideas.

D. Managerial Economies of Scale:

Managerial economies refer to production in managerial costs and proper management of large scale firm. Under this, work is divided and subdivided into different departments. Each department is headed by an expert who keeps a vigil on the minute details of his department.

A small firm cannot afford this specialisation. Experts are able to reduce the costs of production under their supervision. These also arise due to specialization of management and mechanisation of managerial functions.

E. Economies of Transport and Storage:

A firm producing on large scale enjoys the economies of transport and storage. A big firm can have its own means of transportation to carry finished as well as raw material from one place to another. Moreover, big firms also enjoy the economies of storage facilities. The big firm also has its own storage and go down facilities. Therefore, these firms can store their products when prices are unfavorable in the market.

EXTERNAL ECONOMIES:

□ External economies refer to all those benefits which accrue to all the firms operating in a given industry. Generally, these economies accrue due to the expansion of industry and other facilities expanded by the Government.

□ According to Cairncross, “External economies are those benefits which are shared in by a number of firms or industries when the scale of production in any industry increases.” Moreover, the simplest case of an external economy arises when the scale of production function of a firm contains as an implicit variable the output of the industry. A good example is that of coal mines in a locality.

□ Occur outside the organization. These economies occur within the industries which benefit organizations. When an industry expands, organizations may benefit from better transportation network, infrastructure, and other facilities. This helps in decreasing the cost of an organization.

A. Economies of Concentration:

As the number of firms in an area increases each firm enjoys some benefits like, transport and communication, availability of raw materials, research and invention etc. Further, financial assistance from banks and non-bank institutions easily accrue to firm. We can, therefore, conclude that concentration of industries lead to economies of concentration.

B. Economies of Information:

When the number of firms in an industry expands they become mutually dependent on each other. In other words, they do not feel the need of independent research on individual basis. Many scientific and trade journals are published. These journals provide information to all the firms which relates to new markets, sources of raw materials, latest techniques of production etc.

C. Economies of Disintegration

As an industry develops, all the firms engaged in it decide to divide and sub-divide the process of production among themselves. Each firm specializes in its own process. For instance, in case of moped industry, some firms specialize in rims, hubs and still others in chains, pedals, tires etc. It is of two types-horizontal disintegration and vertical disintegration. In case of horizontal disintegration each firm in the industry tries to specialize in one particular item whereas, under vertical disintegration every firm endeavors to specialize in different types of items. Material of one firm may be available and useable as raw materials in the other firms. Thus, wastes are converted into by-products.

The selling firms reduce their costs of production by realizing something for their wastes. The buying firms gain by getting other firms' wastes as raw materials at cheaper rates. As a result of this, the average cost of production declines.

SIGNIFICANCE OF ECONOMIES OF SCALE:

The significance of economies of scale is discussed as under:

(a). Nature of the Industry:

The foremost significance of economies of scale is that it plays an important role in determining the nature of the industry i.e. increasing cost industry, constant cost industry or decreasing cost industry.

(b). Analysis of Cost of Production:

When an industry expands in response to an increase in demand for its products, it experiences some external economies as well as some external diseconomies. The external economies tend to reduce the costs of production and thereby causing an upward shift in the long period average cost curve, whereas the external diseconomies tend to raise the costs and thereby causing an upward shift in the long period average cost curve. If external diseconomies outweigh the external economies, that is, when there are net external diseconomies, the industry would be an Increasing cost industry.

Diseconomies of scale are when the cost per unit of production (Average cost) increases because the output (sales) increases. **Reasons for diseconomies of scale**

1. Communication - becomes more complex
2. Coordination - between departments
3. X- Inefficiency - management costs increase (non-productive costs)
4. Principle agent problem - delegating to employees who are not as committed as the owner

Cost analysis:

Cost Analysis refers to the measure of the cost – output relationship, i.e. the economists are concerned with determining the cost incurred in hiring the inputs and how well these can be re- arranged to increase the productivity (output) of the firm.

Cost refers to the money value that is incurred in acquiring the resources and producing the product.

Determinants of Cost:

Level of output: The cost of production varies according to the quantum of output. If the size of production is large then the cost of production will also be more.

Price of input factors: A rise in the cost of input factors will increase the total cost of production.

Productivities of factors of production: When the productivity of the input factors is high then the cost of production will fall.

Size of plant: The cost of production will be low in large plants due to mass production with mechanization.

Time period: In the long run cost of production will be stable.

Technology: When the organization follows advanced technology in their process then the cost of production will be low.

TYPES OF COST

Fixed Costs (FC): The costs which don't vary with changing output. Fixed costs might include the cost of building a factory, insurance and legal bills. Even if your output changes or you don't produce anything, your fixed costs stay the same.

Costs (VC): Costs which depend on the output produced. For example, if you produce more cars, you have to use more raw materials such as metal. This is a variable cost.

Marginal Costs (MC): is the cost that incurred due to the production of an extra or additional unit.

Opportunity Cost: Opportunity cost is the next best alternative foregone. "It is cost incurred due to one's own choice", i.e the opportunity an individual loose by choosing other.

Incremental cost: Incremental cost is the total **cost** incurred due to an additional unit of product being produced.

Explicit costs: these are costs that a firm directly pays for and can be seen on the accounting sheet. Explicit costs can be variable or fixed and it is a clear amount.

Implicit costs: these are opportunity costs, which do not necessarily appear on its balance sheet but affect the firm. For example the assets of the owner used for the purpose of business.

Sunk Cost: these are costs that have been incurred and cannot be recouped. Depreciation is considered as one of the best example for sunk cost. Example if you spend money on advertising to enter an industry, you can never claim these costs back.

Future Cost: It is a type of cost, where the cost incurred to estimate the future data or forecasting of the organization. It is a type of cost which cannot be exceptional in nature.

Social Cost: This is the total cost to society. It will include the private costs plus also the external cost (cost incurred by a third party). May also be referred to as „True costs“ where the company spends on betterment of the society.

COST CURVES:

Total fixed costs: Given that *total fixed costs* (TFC) are constant as output increases, the curve is a horizontal line on the cost graph.

In economic analysis, the following types of costs are considered in studying cost data of a firm:

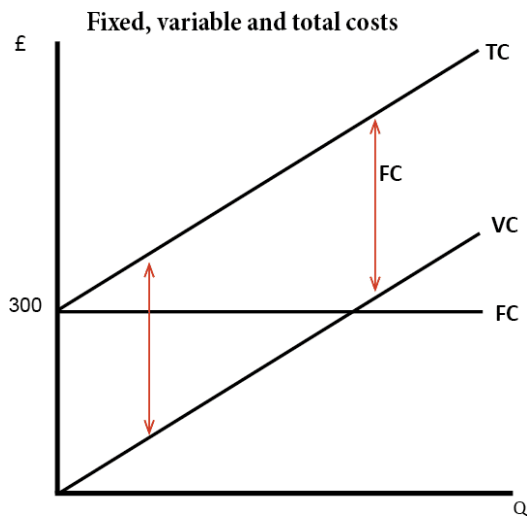
- ☐ Total Cost (TC),
- ☐ Total Fixed Cost (TFC),
- ☐ Total Variable Cost (TVC),
- ☐ Average Fixed Cost (AFC),
- ☐ Average Variable Cost (AVC),
- ☐ Average Total Cost (ATC), and
- ☐ Marginal Cost (MC).

Total Fixed Cost (TFC). It is the cost pertaining to all fixed inputs like machinery, etc., at any given level of output.

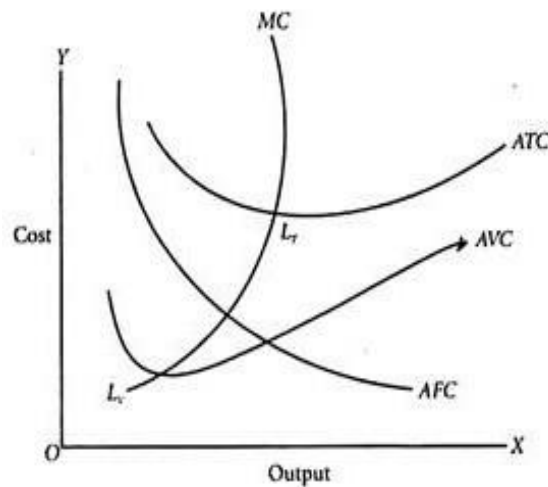
☐ **Total Variable Cost (TVC).** It is the cost pertaining to all variable inputs like raw materials, etc., at any given level of output.

Total Cost (TC). It is the cost pertaining to the entire factor inputs at any given level of output. It is the total cost of production derived by aggregating total fixed and variable costs together.

Thus, $TC = TFC + TVC$.



Short-run Output Cost Curves



Units of Output Q	Total fixed cost TFC	Total variable cost TVC	Total cost (TFC + TVC) TC	Average variable cost (TVC / Q) AVC	Average fixed cost (TFC / Q) AFC	Average cost (TC/Q) AC	Marginal cost MC
0	60	—	60	—	—	—	—
1	60	20	80	20	60	80	20
2	60	36	96	18	30	48	16
3	60	48	108	16	20	36	12
4	60	64	124	16	15	31	16
5	60	90	150	18	12	30	26
6	60	132	192	22	10	32	42

Behaviour of Total Costs

Examining cost schedules in Table 9.1, we may observe the following interesting points about the behaviour of various total costs:

- **TFC remains constant at all levels of output.** It is the same even when the output is nil. Fixed costs are thus independent of output.
- **TVC varies with the output.** It is nil when there is no output. Variable costs are, thus, direct costs of the output.

TVC does not change in the same proportion. Initially, it is increasing at a decreasing rate, but after a point, it increases at an increasing rate. This is due to the operation of the law of variable proportions or non-proportional output, which suggests that initially to obtain a

given amount of output relatively, variations in factors are needed in less proportion, but after a point when the diminishing phase operates, variable factors, are to be employed in a greater proportion to increase the same „level“ of output.

□ **TC varies in the same proportion as the TVC.** Thus, in the short period, the changes in total cost are entirely due to changes in the total

THE BEHAVIOUR OF SHORT-RUN AVERAGE COST CURVES

The behaviour patterns and relations of short-run unit costs become more explicit when we plot the cost data on a graph and draw the respective cost curves.

Average Fixed Cost in Short Run: The average fixed cost is the total fixed cost divided by the volume of output. There is an inverse relation between output and average fixed cost. With the increase in output average fixed cost decreases and with the decrease in output the average fixed cost will increase.

It is calculated from the following formula:

$$AFC = TFC/Q$$

- Q is volume of output
- AFC and TFC are average fixed cost and total fixed cost.

2. Average Variable Cost (AVC): The average variable cost is total variable cost divided by the volume of output. Average variable cost falls with the increase in output, reaches at its minimum and then starts rising. By the operation of law of increasing returns the AVC decreases, and by the operation of constant returns leads to constancy in AVC and the law of diminishing returns leads to increase in AVC. The shape of AVC is U-shaped because of the operation of the laws of returns during short period.

The AVC is calculated by the formula given below:

$$AVC = TVC/Q$$

AVC and TVC are average variable cost and total variable cost while Q is the volume of output.

3. Average Total Cost (ATC): Average Total Cost (ATC) is the aggregate of AFC and AVC. ATC can be calculated from total cost (TC) divided by the volume of output or by aggregating AVC and AFC.

The ATC curve decreases with the increase in output and remains constant up to a point and thereafter it increases with the increase in output. Its shape is U-shaped because of the operation of the laws of return during short period.

The following is the formula of calculating AC:

$$ATC = TC/Q \text{ or } ATC = AFC + AVC$$

ATC and TC are average total cost and total cost while Q is the volume of output.

4. Marginal Cost (MC): It is an addition to total cost by producing an additional unit of output. It can be calculated as the change in total cost divided by an additional unit change in the output.

Marginal Cost changes with the change in average variable cost (AVC) and it is independent of average fixed cost (AFC). Initially, MC Falls, reaches the minimum and thereafter continuously increases. MC is also U-shaped curve. MC curve cuts the ATC and AVC curves at their minimum points.

$$MC = \text{Change in ATC} / \text{Change in Q}$$

****Note:**

- (1) AFC declines continuously, approaching both axes asymptotically (as shown by the decreasing distance between ATC and AVC) and is a rectangular hyperbola.
- (2) AVC first declines, reaches a minimum at Q₂ and rises thereafter. When AVC is at its minimum, MC equals AVC.
- (3) ATC first declines, reaches a minimum at Q₃, and rises thereafter. When ATC is at its minimum, MC equals ATC.
- (4) MC first declines, reaches a minimum at Q₁, and rises thereafter. MC equals both AVC and ATC when these curves are at their minimum values.

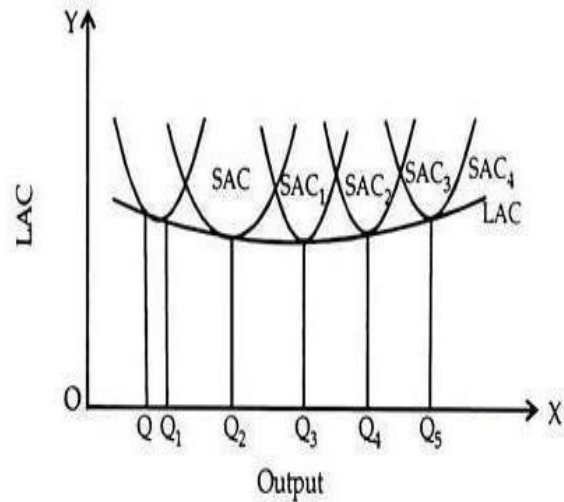
Cost – Output Relationship in Long Run:

Long period gives sufficient time to business managers to change even the scale of production. All the factors of production are variable. All the costs are variable costs and there is no fixed cost. The supply of goods can be adjusted to their demands because scale of production and factors of production can be changed.

Log Run Average Cost (LAC): In the long run, all the factors of production are variable and the firm has a variety of choices to select the size of the plants and the factors of production to be employed. Various short run average cost curves represent the various sizes of the plants available to a firm. We can get the long run average cost curve with the help of all the short run

average cost curves. The long run average cost curve envelopes all the short run average cost curves in it. It is also called an 'Envelope Curve' or 'Planning Curve'.

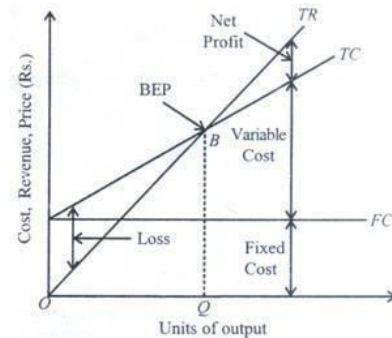
From the above figure long run cost represented on OY-axis and output on OX-axis. SAC, SAC₁, SAC₂, SAC₃ and SAC₄ are short run average cost curves which represent the different size of plants. LAC has been drawn by combining all those points of least cost of producing the corresponding output. The least per unit cost of production is OQ, OQ₁, OQ₂, OQ₃, OQ₄, and OQ₅ respectively.



BREAK-EVEN ANALYSIS

The *BEA* is an important technique to trace the relationship between costs, revenue and profits at the varying levels of output or sales

In *BEA*, the break-even point is located at that level of output or sales at which the net income or profit is zero. At this point, total cost is equal to total revenue. Hence, the breakeven point is the no-profit-no-loss zone.



However, the object of the *BEA* is not just to determine the break-even point (*BEP*), but to understand the functional relationship among cost, revenue and the rate of output.

ASSUMPTIONS OF BREAK-EVEN ANALYSIS

The validity of the *BEA* is conditioned by a number of assumptions as follows:

1. The cost function and the revenue function are linear.
2. The total cost is divided into fixed and variable costs.
3. The selling price is constant.
4. The volume of sales and the volume of production are identical.
5. Average and marginal productivity of factors are constant.
6. The product-mix is stable in the case of a multi-product firm.
7. Factor price is constant.

LIMITATIONS OF BEA

The break-even analysis has certain major limitations as follows:

- **It is static.** In the *BEA*, everything is assumed to be constant. This implies a static condition. It is not suited to a dynamic situation.
- **It is unrealistic.** It is based on many assumptions which do not hold good in practice. Linearity of cost and revenue function are true only for a limited range of output.
- **It has many shortcomings.** The *BEA* regards profit as a function of output only. It fails to consider the impact of technological change, better management, division of labour, improved productivity and such other factors influencing profits.
- **Its scope is limited to the short-run only.** The *BEA* is not an effective tool for a long-run analysis.
- **It assumes horizontal demand curve with the given price of the product.** But this is not so in the case of a monopoly firm.
- **It is difficult to handle selling costs in the BEA.** Selling costs do not vary with output. They manipulate sales and affect the volume of output.
- **The traditional BEA is very simple.** It makes no provision for corporate income-tax, etc.

Managerial Use of Break Even Analysis:

(i) **Safety Margin:** The break-even chart helps the management to know at a glance the profits generated at the various levels of sales. The safety margin refers to the extent to which the firm can afford a decline before it starts incurring losses.

The formula to determine the sales safety margin is:

$$\text{Safety Margin} = (\text{Sales} - \text{BEP}) / \text{Sales} \times 100$$

(ii) **Target Profit:** The break-even analysis can be utilised for the purpose of calculating the volume of sales necessary to achieve a target profit.

(iii) **Change in Price:** The management is often faced with a problem of whether to reduce prices or not. Before taking a decision on this question, the management will have to consider a profit. A reduction in price leads to a reduction in the contribution margin.

This means that the volume of sales will have to be increased even to maintain the previous level of profit. The higher the reduction in the contribution margin, the higher is the increase in sales needed to ensure the previous profit.

(iv) Change in Costs:

When costs undergo change, the selling price and the quantity produced and sold also undergo changes.

Changes in cost can be in two ways:

(i) Change in variable cost, and

(ii) Change in fixed cost.

(i) Variable Cost Change: An increase in variable costs leads to a reduction in the contribution margin. This reduction in the contribution margin will shift the break-even point downward. Conversely, with the fall in the proportion of variable costs, contribution margins increase and break-even point moves upwards.

(ii) Fixed Cost Change: An increase in fixed cost of a firm may be caused either due to a tax on assets or due to an increase in remuneration of management, etc. It will increase the contribution margin and thus push the break-even point upwards. Again to maintain the earlier level of profits, a new level of sales volume or new price has to be found out.

(v) Decision on Choice of Technique of Production: The breakeven analysis is the most simple and helpful in the case of decision on a choice of technique of production. For low levels of output, some conventional methods may be most probable as they require minimum fixed cost. For high levels of output, only automatic machines may be most profitable. By showing the cost of different alternative techniques at different levels of output, the break-even analysis helps the decision of the choice among these techniques.

(vi) Make or Buy Decision: Firms often have the option of making certain components or for purchasing them from outside the concern. Break-even analysis can enable the firm to decide whether to make or buy.

(vii) Plant Expansion Decisions: The break-even analysis may be adopted to reveal the effect of an actual or proposed change in operation condition. This may be illustrated by showing the impact of a proposed plant on expansion on costs, volume and profits. Through the break-even analysis, it would be possible to examine the various implications of this proposal.

(viii) Plant Shut Down Decisions: In the shut down decisions, a distinction should be made between out of pocket and sunk costs. Out of pocket costs include all the variable costs plus the fixed cost which do not vary with output. Sunk fixed costs are the expenditures previously made but from which benefits still remain to be obtained e.g. depreciation.

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(ix) Advertising and Promotion Mix Decisions: The break-even point concept helps the management to know about the circumstances. It enables him not only to take appropriate decision but by showing how these additional fixed cost would influence BEPs. The advertisement pushes up the total cost curve by the amount of advertisement expenditure.

(x) Decision Regarding Addition or Deletion of Product Line: If a product has outlive utility in the market immediately, the production must be abandoned by the management and examined what would be its consequent effect on revenue and cost. Alternatively, the management may like to add a product to its existing product line because it expects the product as a potential profit spinner. The break-even analysis helps in such a decision.

Practical problems on break even analysis